

Class #21 - Null Spaces and Column Spaces

1. Def: The *null space* of an $m \times n$ matrix A , written as $\text{Nul } A$, is the set of all solutions to the _____ equation $Ax = 0$, i.e.,

$$\text{Nul } A = \{x \in \mathbf{R}^n : \text{_____}\}$$

Note: $\text{Nul } A$ is a _____ of _____.

2. Example: Let H be the set of all vectors in $\mathbf{R}^4$ whose coordinates a, b, c, d satisfy the equation $a - 2b + 5c = d$ and $c - a = b$. Show that H is a subspace of $\mathbf{R}^4$.

3. Example: Find a spanning set for the $\text{Nul } A$:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

It can be shown that $A \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

Note:

- (a) The spanning set produced this way is _____ if A has a _____ column (that is, there is a free variable).

If every column of A is a pivot column, then $\text{Nul } A =$ _____, and the spanning set is _____.

- (b) If A has a nonpivot column, then

the number of vectors in the spanning set for $\text{Nul } A =$ _____.

4. Def: The *column space* of an $m \times n$ matrix A , written as $\text{Col } A$, is the set of all linear combinations of the _____ of A , i.e.,

$$\text{Col } A = \text{Span}\{a_1, a_2, \dots, a_n\}.$$

Note:

- (a) $\text{Col } A$ is a _____ of _____, and $\text{Nul } A$ is a _____ of _____.

- (b) $\text{Col } A = \{b \in \text{_____} : Ax = b \text{_____}\}$

- (c) $\text{Col } A = \text{Range } \text{_____}$

- (d) $\text{Col } A = \mathbf{R}^m$ if and only if $Ax = b$ _____.

5. Def: A transformation T from a vector space V into a vector space W is a *linear transformation* if

- (i) _____ for _____

- (ii) _____ for _____

6. Example: Let V be the set of all real-valued continuous functions whose derivative is continuous on the interval $[a, b]$. (V is a _____.)
- Let W be the set of all real-valued continuous functions on $[a, b]$. (W is a _____.)
- Let D be a transformation from V to W such that $D(f(t)) = f'(t)$.
- Is D linear?
7. Def: The *kernel* (or *null space*) of a linear transformation T is the set of all $u \in$ _____ such that $T(u) = 0$.
8. If $T : \mathbf{R}^n \rightarrow \mathbf{R}^m$ is linear, then
- Nul _____ = Ker T , and Col _____ = _____ T (where A is the _____ of T).
9. Example: What is the Ker D of #6?

10. Example: Let V be the set of all real-valued continuous functions whose 1st and 2nd order derivatives are continuous on the interval $[a, b]$. (V is a _____.)
- Let W be the set of all real-valued continuous functions on $[a, b]$. (W is a _____.)
- Let $L : V \rightarrow W$ be the transformation such that $L(y(t)) = y''(t) + cy(t)$. (L is _____.)
- The solution set of the differential equation $y'' + cy = 0$ is _____ of L , and $\text{Ker } L$ is a _____ of _____.